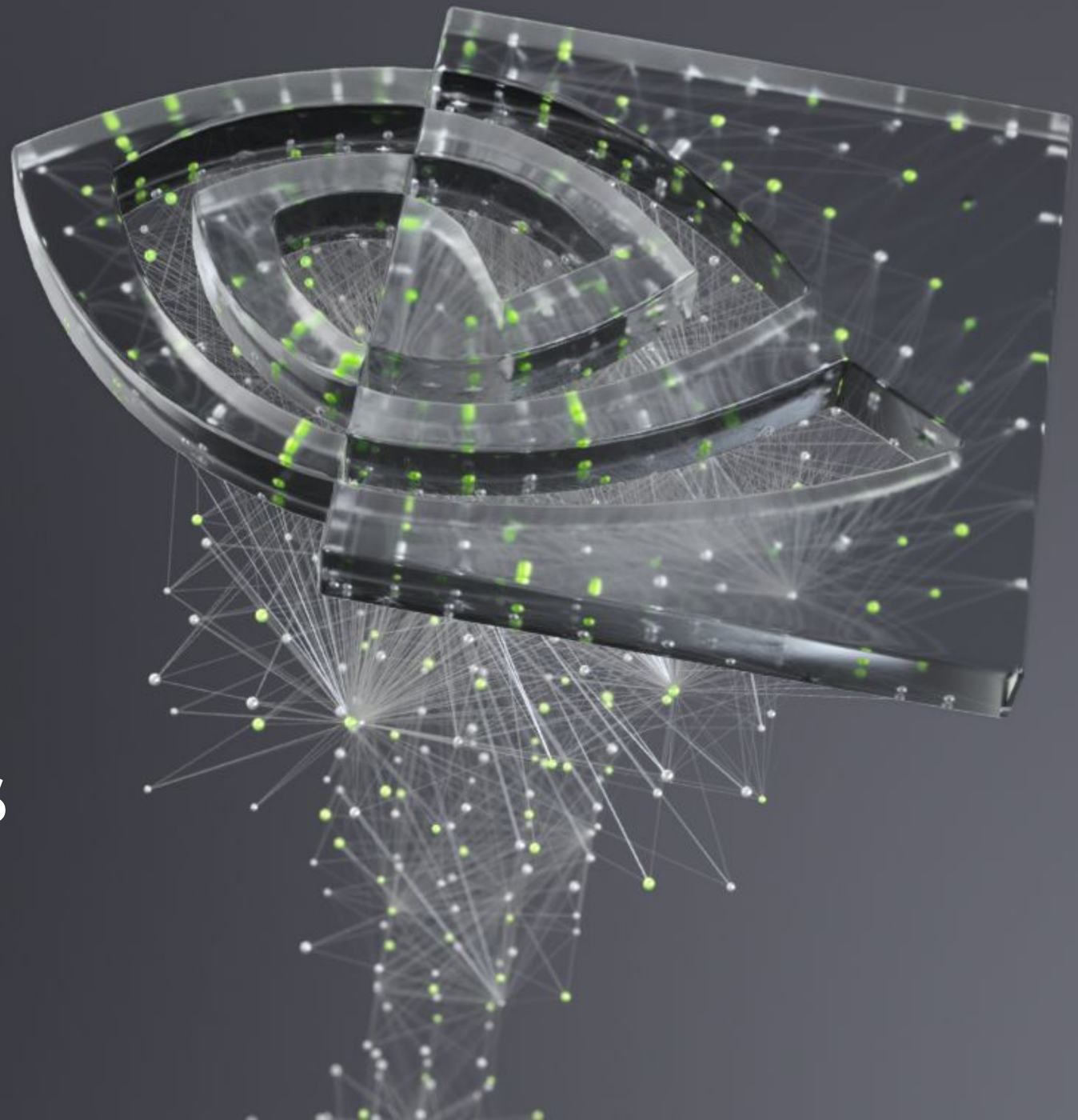


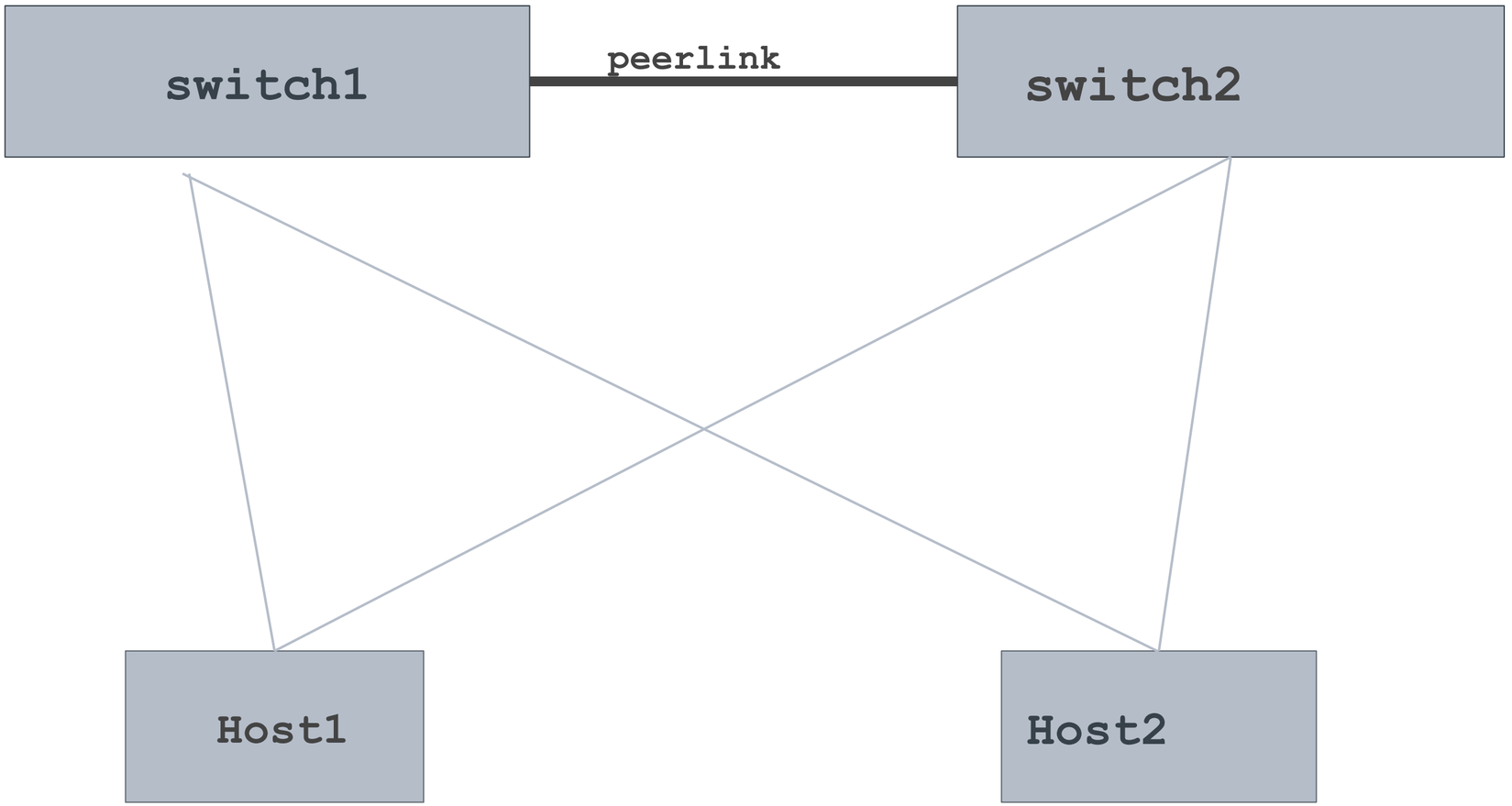
Networking Updates

Roopa Prabhu

Aug 14, 2020

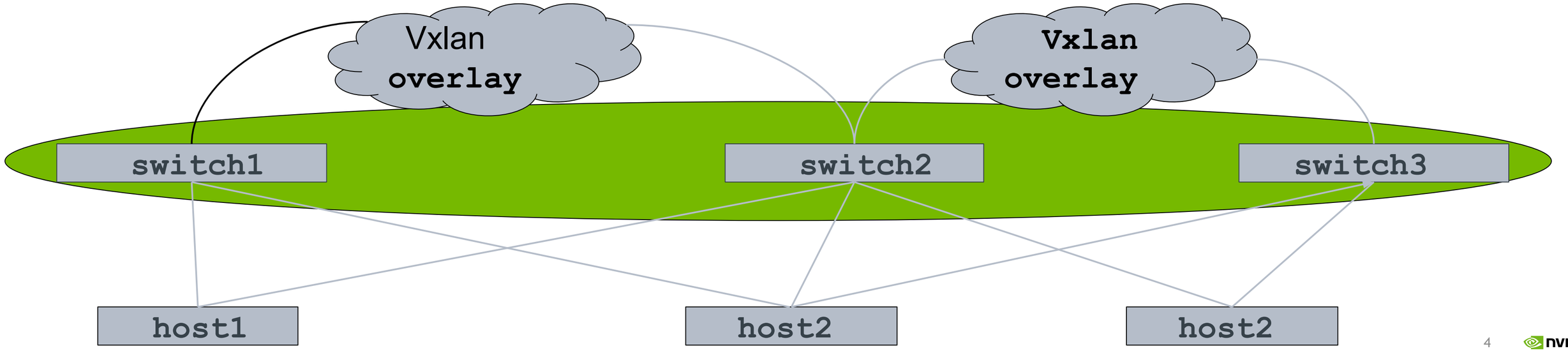
Linux Kernel dataplane for an Open standards based multihoming protocol

Traditional Multihoming



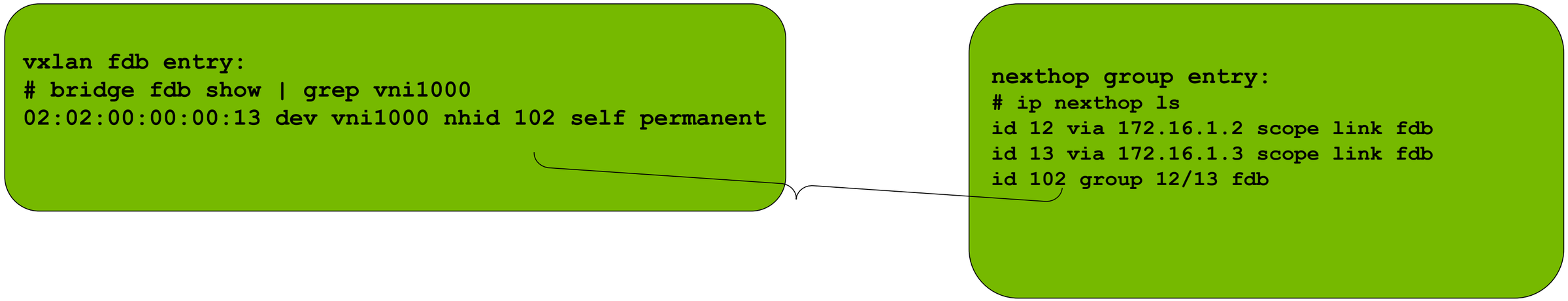
Open Multihoming solution with VxLAN Overlay

- E-VPN multihoming: BGP based E-VPN multihoming controlplane [1]
  - Connect your servers to a redundant pair of switches running Open BGP based multihoming protocol
  - Peer switches are connected over VxLan overlay (peer switches are vxlan tunnel endpoints or VTEPs)

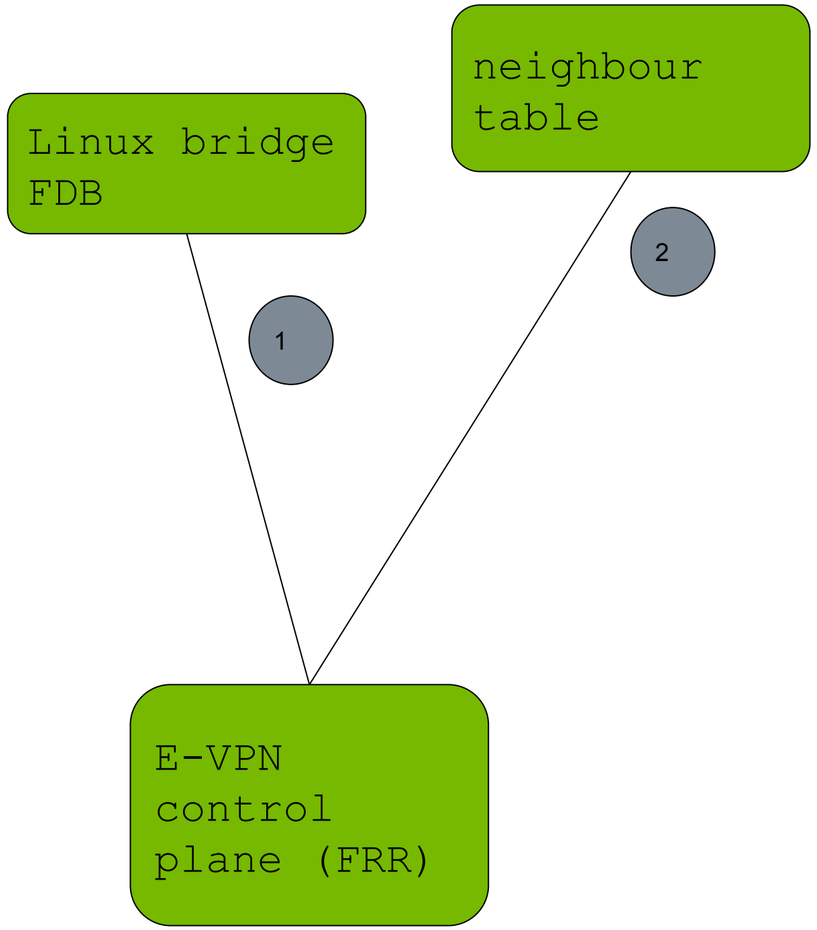


Linux Kernel Dataplane forwarding enhancements to support E-VPN-multihoming

- Vxlan FDB ECMP nexthop groups support [2]
  - Ability to ECMP to multiple evpn peered vteps



Encoding local vs peer ownership in neigh entries



- Kernel FDB and neighbor database is central to a Multihoming protocol
- Keeping them in sync across multihoming peers for faster convergence is key
- Kernel API enhancements for accuracy and better convergence amidst mac moves in these systems (With requests from FRR team):

1 Bridge notify: To indicate a MAC has become active locally due to kernel dataplane seeing a packet on a host port locally [3]

2 Neighbor entry enhancements to indicate local reachability and multihoming-peer reachability. new flag (pending upstream)

Miscellaneous updates

Protodown and protodown reason

- protodown is a per netdevice flag today that enables control plane protocols to hold an interface carrier down
- Multiple users:
  - Multihoming protocols
  - VRRP
  - port security violation
  - flaky link, auto-detect and keep the link down
- New protodown-reason support upstream. (iproute2 changes pending)

```
$cat /etc/iproute2/protodown_reasons.d/r.conf
0 mlag
1 evpn
2 vrrp
3 psecurity

$ip link set dev vxlan0 protodown on protodown_reason vrrp on
$ip link set dev vxlan0 protodown_reason mlag on
$ip link show
14: vxlan0: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode
DEFAULT group default qlen 1000
    link/ether f6:06:be:17:91:e7 brd ff:ff:ff:ff:ff:ff protodown on <mlag,vrrp>
```

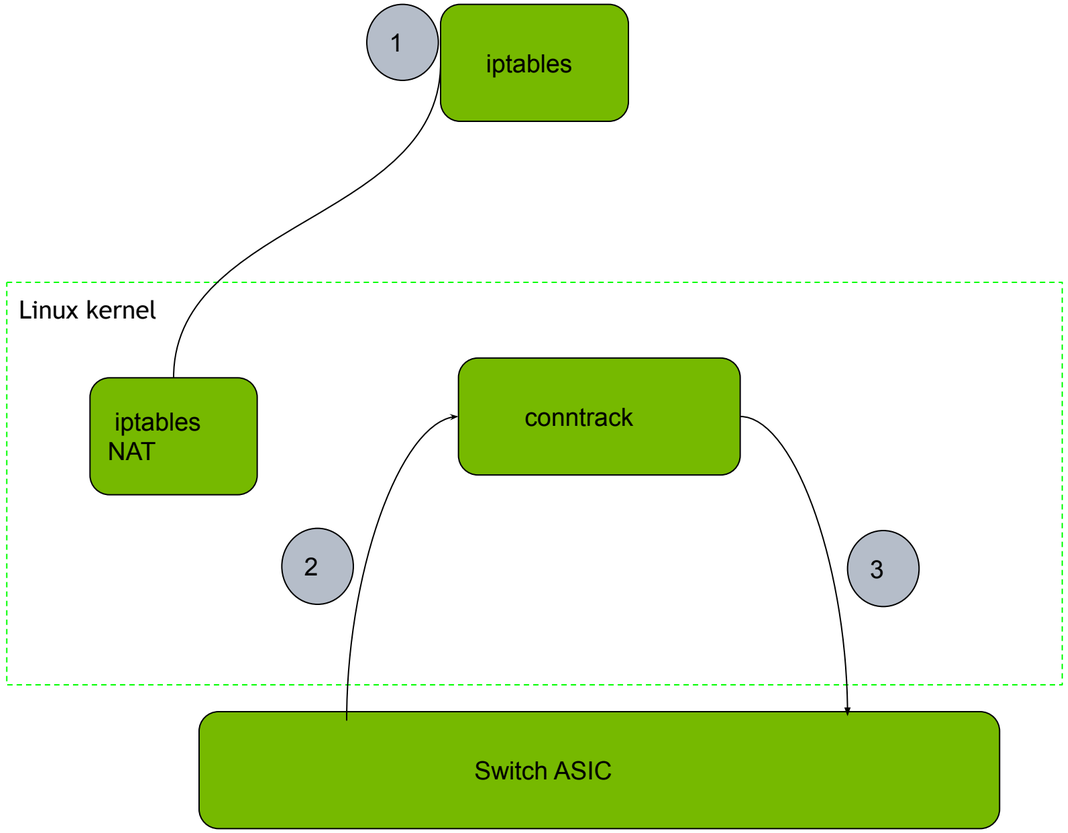
NAT offload to switch ASIC

- Move NAT function from host to the ASICs on the switch: NAT at higher speeds and scale
- Linux kernel NAT offload to switch ASIC: iptables/nftables/conntrack or TC conntrack

Dynamic NAT offload:

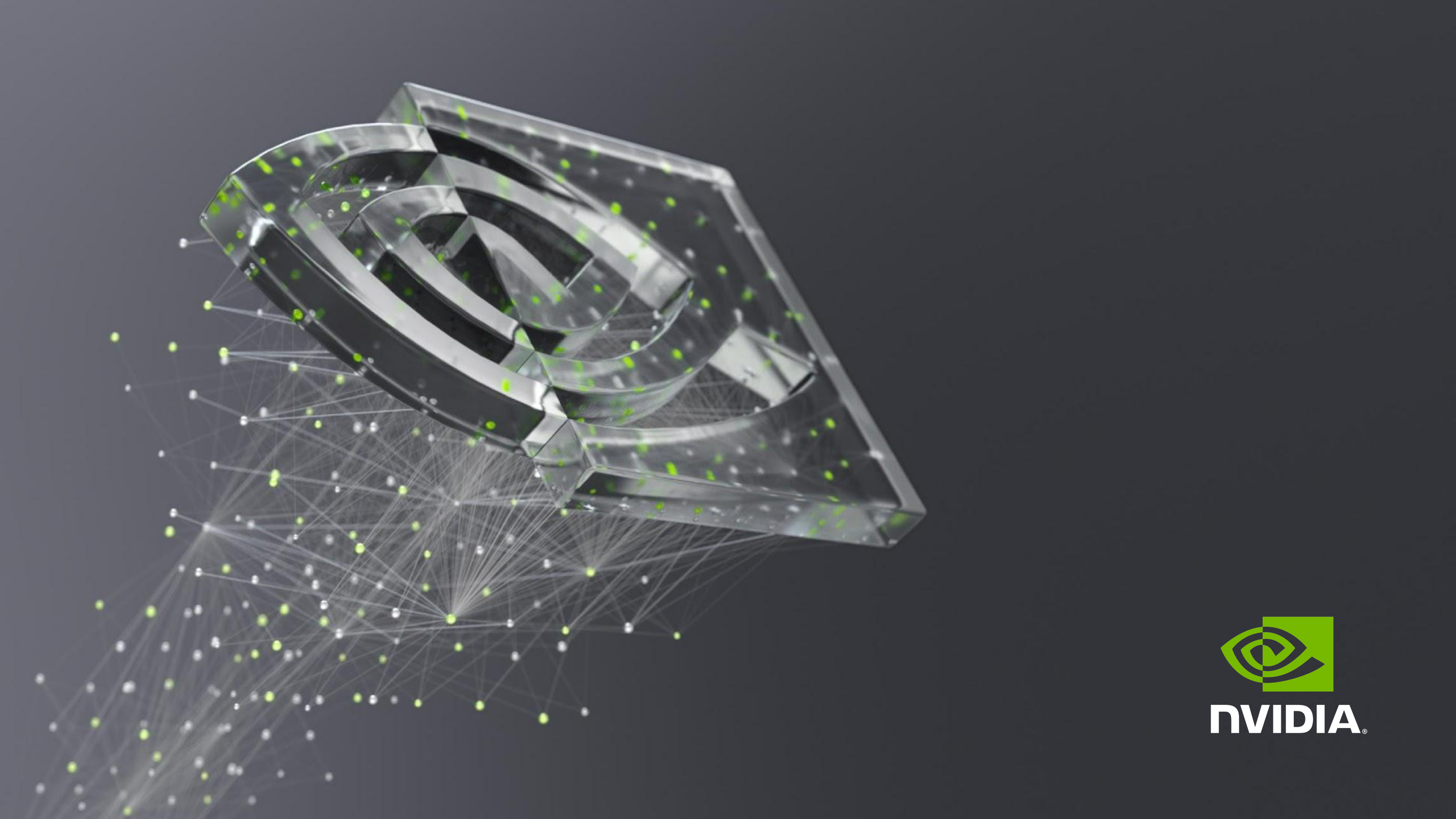
1. iptables dynamic NAT entry
2. Trap first packet to CPU matching NAT rule
3. On conntrack entry learn, ofload conntrack entry to HW

   (Offload via netlink or in-kernel offload API)



References

[1] E-VPN Multihoming: https://tools.ietf.org/html/rfc7432#section-8
[2] VxLAN FDB nexthop groups:
https://patchwork.ozlabs.org/project/netdev/cover/1590125177-39176-1-git-send-email-roopa@cumulusnetworks.com/
[3] Bridge notify flag:
https://patchwork.ozlabs.org/project/netdev/cover/20200623204718.1057508-1-nikolay@cumulusnetworks.com/
[4] Protodown reason:
https://patchwork.ozlabs.org/project/netdev/patch/1596242041-14347-1-git-send-email-roopa@cumulusnetworks.com/
[5] NAT offload on Cumulus Linux:
https://docs.cumulusnetworks.com/cumulus-linux-41/Layer-3/Network-Address-Translation-NAT/


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